Urban Mobility in Seoul: Predicting Bike-Sharing Demand

Abstract

This project predicts hourly bike rental demand in Seoul using temporal and environmental features from the UCI Seoul Bike Sharing dataset. A machine learning pipeline is developed with feature engineering techniques including cyclical time encoding and lag-based demand features.

Multiple models are evaluated, including linear regression, a feedforward neural network, and gradient boosted decision trees (XGBoost). XGBoost achieves the strongest performance ($R^2 = 0.94$ on the test set), demonstrating the effectiveness of tree-based methods for structured time series prediction.

Introduction

Seoul's bike-sharing program (Ddareungi) has expanded from 2,000 bikes in 2015 to over 45,000 in 2024, supporting more than 200 million rides. As the system scales, accurate demand forecasting becomes critical to reduce shortages and inform infrastructure planning.

This project models hourly bike demand using temporal and environmental features, including time of day, seasonality, and weather conditions.

Related Work

Prior work demonstrates that bike demand in Seoul is predictable using temporal and weather features. Satishkumar and Cho (2020) evaluate multiple algorithms: CUBIST, regularized random forest, CART, KNN, and conditional inference trees, finding that a CUBIST (a rule-based model) achieved the best performance. Their work highlights the importance of model comparisons and temporal feature transformations.

Yuan et. al (2024) focus on interpretability and causal relationships using linear regression models with progressively added feature groups. Their best model achieves $R^2 = 0.6$ and identifies rainfall as the most influential variable.

Building on these works, this project extends beyond linear approaches by incorporating nonlinear models (XGBoost, neural networks) and additional temporal feature engineering (cyclical encoding, lag features) to improve performance.

Dataset

This dataset is sourced from the UCI Machine Learning Repository and contains hourly observations from 2017–2018.

- Initial size: 8,760 observations, 13 features

- Final dataset: ~8,400 observations after preprocessing
- Target: hourly bike rental count
- Temporal features: hour, season, day of year
- Environmental features: temperature, humidity, wind speed, visibility, rainfall, snowfall

Preprocessing

Irrelevant and highly collinear features were removed to reduce redundancy. The dataset was filtered to include only operational periods ("functioning day = yes"). The date field was converted to datetime format, and a day-of-year feature was derived.

Categorical variables (Holiday, Season) were one-hot encoded. To capture periodic patterns, cyclical encoding was applied to hour and day of year using sine and cosine transformations, after which the original variables were removed.

To incorporate temporal dependencies, lag-based features were introduced: 1 hour, 24 hour, and 3-hour rolling mean. These features capture short-term trends and daily recurrence patterns.

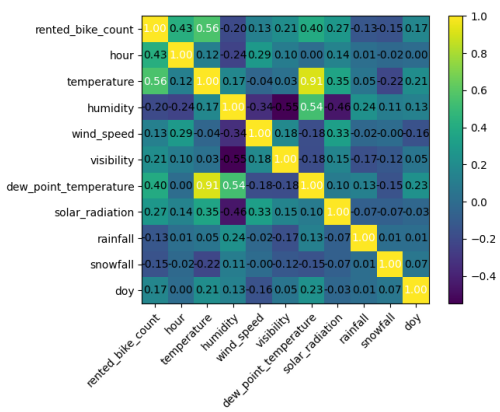
Numeric features were scaled using MinMax normalization. The dataset was split chronologically into training (60%), validation (20%), and test (20%) sets, with scaling fit only on the training data to prevent leakage.

Exploratory Data Analysis

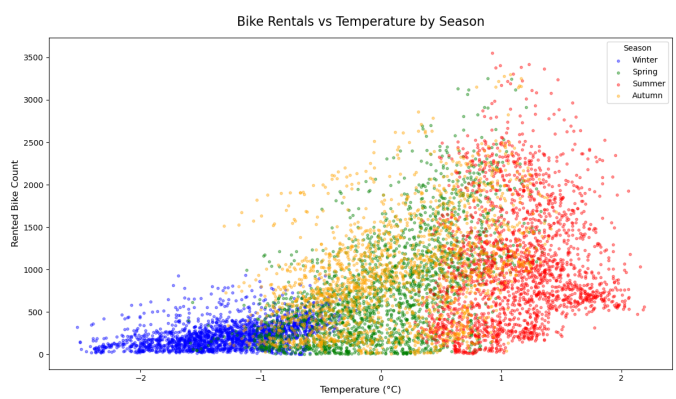
Correlation analysis shows that bike demand is moderately correlated with temperature (0.56), hour (0.43), and dew point temperature (0.40). Temperature and dew point temperature are highly collinear (0.91), leading to removal of the dew point feature.

A scatterplot of demand vs. temperature reveals a clear positive relationship across seasons, with higher variability during warmer months.

Correlation Matrix (Numeric Variables)



Bike Rentals vs. Temperature by Season



Methodology

A baseline model is established by predicting the mean hourly demand from the training set. Three models are evaluated: linear regression, feedforward neural network, and gradient boosted decision trees. Performance is measured using mean absolute error (MAE), root mean square (RMSE), and R^2.

Linear regression serves as an interpretable model to assess the predictive ability of the feature set. The model estimates coefficients that minimize the sum of squared residuals.

A feedforward neural network (FFNN) is implemented to capture complex non-linear relationships. The architecture consists of fully connected layers with ReLU activations and is trained using backpropagation. Hyperparameters (layer size, depth, learning rate, dropout) are tuned using Keras Tuner with random search. The optimal configuration includes two hidden layers (96 and 128 units) with a learning rate of 0.0005. Early stopping and dropout are applied to reduce overfitting.

XGBoost is selected for its strong performance on structured data and ability to model nonlinear feature interactions. The model builds an ensemble of decision trees sequentially using gradient boosting to minimize prediction error. Regularization techniques include tree depth tuning (max_depth = 6) and column subsampling (colsample_bytree = 0.9).

Experiments and Results

Linear Regression:

Linear regression significantly improves over the baseline.

- MAE reduced by 39% (535 vs 327)
- RMSE reduced by 32% (645 vs 438)
- $R^2 = 0.54$ (vs 0)

A log transformation of the target reduces MAE (298) but worsens RMSE (462) and R^2 (0.49). This indicates a tradeoff of improved average error but reduced sensitivity to high demand periods due to compression of extreme values.

Feedforward Neural Network (FFNN):

The FFNN achieves performance comparable to linear regression but does not outperform it. This is likely due to the relatively small dataset and structured feature space, where engineered features already capture much of the predictive signal.

XGBoost:

XGBoost achieves the strongest performance across all evaluation metrics.

- MAE: Validation (118), Test (96)
- RMSE: Validation (198), Test (145)
- R^2: Validation (0.91), Test (0.94)

Despite signs of overfitting, the model generalizes well and effectively captures nonlinear relationships in demand.

Performance Comparison:

Model	MAE	RMSE	R^2
Baseline	~535	~645	~0.00
Linear Regression	327	438	0.54
FFNN	326	446	0.53
XGBoost (Val)	118	198	0.91
XGBoost (Test)	96	145	0.94

Conclusion

XGBoost significantly outperforms all evaluated models, achieving an R^2 of 0.94 on the test set. The results demonstrate that bike demand is highly predictable when incorporating temporal features, particularly lag-based features that capture strong autoregressive patterns.

In terms of limitations and future work, some overfitting remains, as training error is lower than validation error. Future improvements could include more systematic hyperparameter tuning, stronger regularization strategies, and comparison with alternative boosting frameworks such as LightGBM.

References

[1] Satishkumar, V.E., & Cho, Y. (2020). Forecasting bike demand in Seoul...

[2] Yuan, Q., et al. (2024). Modeling bike demand...